LAST NAME: _____

Practice Quiz 07A — MATH 2055 — Fall 2026

Practice for Quiz 07

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

Recall that $u \equiv v \pmod 3$ means $3 \mid (u - v)$, and that $d \mid n$ means $n = dk$ for some integer k.

1. Define a relation on $\mathbf{Z}$ by $x \sim y$ if and only if $x \equiv y \pmod 3$.

   (a) Prove directly from the definition that $\sim$ is transitive.

   (b) Assume that $\sim$ is also reflexive and symmetric. State the number of equivalence classes and list the classes.